

Rotary Wing Aerodynamics

Module 13d • Study cards • 170 cards • Wingman

1. What is the primary airfoil of the helicopter

The main rotor

The manual names the main rotor as the helicopter's primary airfoil.

2. Why is the helicopter often referred to as a rotary-wing aircraft

Its primary airfoil is the main rotor

Because its main lifting surface is a rotating rotor, the helicopter is called rotary-wing.

3. What is an airfoil by definition

Any surface which generates a useful dynamic reaction of the air

The manual defines an airfoil as any surface giving a useful dynamic reaction of the air.

4. Which two forces are named as the useful reaction of the air on an airfoil

Lift and thrust

The text names lift and thrust as the two forces needed for flying and manoeuvring.

5. What are the airfoils of the main rotor

The rotor blades

The blades are the main rotor airfoils.

6. What is the root of a rotor blade

The end where the blade is attached to the rotor mast

The root is the blade end attached to the rotor mast.

7. What is the tip of a rotor blade

The outer end of the blade, opposite the root

The tip is the outer end, opposite the root.

8. What is the span of a rotor blade

The distance from root to tip along the blade centerline

Span is measured root to tip along the centerline.

9. A technician measures a rotor blade from the attachment point at the rotor mast to its outer end, along its centerline. Which dimension is being measured?

The span

Root to tip along the centerline is the span; the chord runs leading edge to trailing edge.

10. What is the leading edge of a rotor blade

The edge that cuts into the wind when the rotor is turning

The leading edge cuts into the wind.

11. What is the trailing edge of a rotor blade

The edge opposite the leading edge

The trailing edge is opposite the leading edge.

12. What is the chord of a rotor blade

A straight line from the leading edge to the trailing edge

The chord is the straight reference line from leading edge to trailing edge.

13. Looking at the cross section of a rotor blade, a straight reference line is drawn from the front edge to the rear edge. What is this line called?

The chord

That reference line in the cross section is the chord.

14. What is the advancing blade

The forward moving blade

The blade moving forward is the advancing blade.

15. What is the retreating blade

The backward moving blade

The blade moving backward is the retreating blade.

16. What is the relative wind

The direction of the airflow with respect to the rotor blade

Relative wind is airflow direction relative to the blade.

17. How is the relative wind directed in relation to the flight path of the blade

Always opposite to the flight path

The relative wind is always opposite to the blade's flight path.

18. A blade moves horizontally forward. In which direction does the relative wind move

Horizontally backward

The relative wind is exactly opposite: horizontally backward.

19. A blade moves forward and downward. In which direction does the relative wind move

Backward and upward

Exactly opposite to forward-and-downward is backward-and-upward.

20. On the retreating side a blade moves horizontally backward. Which relative wind acts on it

Horizontally forward

Relative wind is always exactly opposite to blade movement.

21. A blade is moving forward and at the same time flapping upward. Applying the rule for relative wind, from which direction does the relative wind come?

Backward and downward

The relative wind moves exactly opposite to the blade path, so forward-and-upward blade motion gives a backward-and-downward relative wind.

22. Which two movements of the rotor blades are given as factors affecting the relative wind

Turning and flapping

The manual names turning and flapping of the blades.

23. A helicopter stands with its rotor stopped while wind blows over the blades. Which statement about the relative wind is correct?

The wind blowing over the blade creates the relative wind

Even with the rotor stopped, wind over the blade is airflow relative to the blade.

24. What is the axis of rotation

The axis around which the rotor blades are turning

The rotor blades turn around the axis of rotation.

25. What forms the plane of rotation

The average tip path of the blades

The plane of rotation is formed by the average blade tip path.

26. How is the plane of rotation arranged with respect to the axis of rotation

Perpendicular to it

The plane of rotation is perpendicular to the axis.

27. A helicopter is flying and the blades flap during each revolution, so each tip follows a slightly different path. Which path defines the plane of rotation?

The average tip path of the blades

The manual defines it by the average tip path.

28. What is the pitch angle

The acute angle between the chord and the plane of rotation

Pitch angle is measured from the plane of rotation to the chord.

29. Which cockpit control changes the pitch angle of all main rotor blades

The collective pitch control

The collective pitch control changes the pitch angle.

30. Around which axis does the blade rotate when the pitch angle is changed with the collective pitch control

The hub axis

The manual says the blade rotates around the hub axis.

31. Which term is often confused with the pitch angle

The angle of attack

The note warns that pitch angle is often confused with angle of attack.

32. Which statement correctly distinguishes pitch angle from angle of attack?

Pitch angle is referenced to the plane of rotation, angle of attack to the relative wind

Both use the chord, but pitch angle uses the plane of rotation and angle of attack uses the relative wind.

33. What is the angle of attack

The acute angle between the chord line and the relative wind

Angle of attack is chord line to relative wind.

34. Besides the pitch angle, what can the pilot also vary by operating the collective pitch control?

The angle of attack

The angle of attack can also be varied with the collective.

35. Why do the same factors that affect the relative wind also affect the angle of attack

Because the angle of attack depends upon the relative wind

Angle of attack is measured to the relative wind, so what changes one changes the other.

36. When is the angle of attack equal to the pitch angle

When the relative wind is in the same direction as the plane of rotation

If the relative wind lies in the plane of rotation, both angles use the same reference line.

37. The pilot increases the pitch angle. What happens to the angle of attack

It is also increased

Pitch and angle of attack change together.

38. The pilot lowers the collective and so reduces the pitch angle. What happens to the angle of attack?

It is also decreased

Decreasing pitch also decreases the angle of attack.

39. The pilot holds the pitch angle constant, but outside influences change the relative wind. What happens to the angle of attack?

It changes, because it depends on the relative wind

Angle of attack is also affected by relative wind changes, not only by pitch.

40. How can the pilot increase or decrease the angle of attack

By moving the pitch angle of the rotor blade

The pilot changes the angle of attack through the blade pitch angle.

41. In the figure 'Angle of Attack', the relative wind comes from slightly below the plane of rotation. How does the angle of attack compare with the pitch angle?

The angle of attack is larger than the pitch angle

The figure shows the angle of attack as the larger angle because the relative wind is inclined below the plane of rotation.

42. What is lift

The force that is produced by the airfoil

Lift is the airfoil's force.

43. In which direction does lift act in relation to the relative wind

Perpendicular to the relative wind

Lift is perpendicular to the relative wind.

44. What is lift required to counteract

Gravity

Lift counteracts gravity.

45. To which bodies does gyroscopic precession apply

All rotating bodies

It applies to all rotating bodies, including both rotors.

46. How is gyroscopic precession defined

The change of the rotational axis as a result of an external force

Precession is a change of the rotational axis caused by an external force.

47. Where does the output occur when a force is applied to a turning rotor

90° later in the direction of rotation

The output appears 90° later in the direction of rotation.

48. From which point is the 90° of gyroscopic precession measured

From the point of the input

The 90° is measured from the point of input.

49. In the example wheel, an external force tries to turn the wheel around the input axis. Around which axis does the movement actually take place?

Around the output axis

Because of precession the movement occurs around the output axis.

50. For a helicopter whose rotor turns counterclockwise, where must the downward force be put in to tilt the tip-path plane forward?

On the right side

The manual states the input goes exactly on the right side.

51. In which direction must the tip-path plane be tilted for the helicopter to fly forward

Forward

Forward flight needs a forward-tilted tip-path plane.

52. Why must the steering input to the rotor be applied 90° before the point where the tilt is wanted

Because the output occurs 90° later in the direction of rotation

Precession shifts the result 90° in the direction of rotation, so the input must lead by 90°.

53. A downward force is applied on the right side of a counterclockwise turning rotor. Where does the output appear?

Right in front

90° later in counterclockwise rotation from the right side is the front.

54. A helicopter rotor turns counterclockwise. The tip-path plane must tilt backward for rearward flight. Following the precession rule, where must the downward force be put in?

On the left side

The output is wanted at the rear; 90° earlier in counterclockwise rotation is the left side.

55. A helicopter rotor turns counterclockwise. The tip-path plane must tilt to the left. Following the precession rule, where must the downward force be put in?

Right in front

The output is wanted on the left; 90° earlier in counterclockwise rotation is the front.

56. A downward force is applied at the rear of a counterclockwise turning rotor. Where does the tip-path plane tilt down?

On the right side

90° later than the rear in counterclockwise rotation is the right side.

57. Which law explains torque reaction on a helicopter fuselage

Newton's Third Law

Torque reaction is action and reaction, Newton's Third Law.

58. In which direction does the helicopter fuselage tend to rotate due to torque

Opposite to the direction of the rotor blades

Action and reaction: the fuselage turns opposite to the rotor.

59. What is the effect called when the fuselage tends to rotate opposite to the rotor blades

Torque

The manual calls this effect torque.

60. In a helicopter with a main rotor and a tail rotor, which component compensates the torque

The tail rotor

The tail rotor compensates torque.

61. In the most common helicopter design, what does the main rotor create

Lift and thrust

The main rotor creates lift and thrust.

62. A helicopter's main rotor turns counterclockwise seen from above and the tail rotor is not yet producing thrust. Which way does the fuselage tend to turn?

Clockwise

The fuselage reaction is opposite to a counterclockwise rotor, so clockwise.

63. Why does a helicopter need no pull or push propellers

Because the rotors generate both lift and thrust

The rotor provides both lift and thrust.

64. How do the rotors of a helicopter additionally ensure directional control

The rotor plane may be tilted so thrust can act in all directions

Tilting the rotor plane lets thrust act in any direction.

65. What does lift counteract according to the helicopter description

Weight

Lift counteracts weight.

66. What generates the total rotor thrust

All rotor blades together

The note says all blades together generate total rotor thrust.

67. After an engine failure or broken drive shafts, how can a helicopter land safely

By performing an autorotation

Autorotation allows a safe landing without engine power.

68. During an autorotation after an engine failure, which statement about control is correct?

The helicopter is fully controllable and the pilot can choose the touchdown point

The manual says it is fully controllable and the pilot selects the touchdown location.

69. In the figure 'Helicopter', the total rotor thrust is split into two components. What are they called?

Lift and thrust

The figure shows total rotor thrust with its lift and thrust components.

70. How do helicopters change direction, unlike other aircraft

By changing the angle of the rotor blade plane

Helicopters tilt the rotor blade plane instead of deflecting surfaces.

71. What achieves the change of the rotor blade plane angle for directional control

Mechanical actuators controlled by the pilot and aerodynamic forces on the blades

The manual names pilot-controlled mechanical actuators plus aerodynamic forces.

72. The pilot moves the cyclic stick forward. What does the rotor plane do

It tilts forward

Forward cyclic tilts the rotor plane forward.

73. The pilot moves the cyclic stick to the left. In which direction does the helicopter fly

To the left

Left cyclic gives flight to the left.

74. In which direction does rotor thrust always act

Perpendicular to the rotor plane

Rotor thrust is perpendicular to the rotor plane.

75. Why must the rotor plane be tilted when changing from hover to forward flight

Because rotor thrust always acts perpendicular to the rotor plane

To point thrust forward, the plane must tilt, because thrust is perpendicular to it.

76. What is the technical solution for tilting the rotor plane

Changing the pitch angle during each revolution

Cyclic pitch changes during each revolution tilt the plane.

77. Which movements do helicopters with one main rotor control with the principle of cyclic control?

Longitudinal and lateral movement

Cyclic control handles longitudinal and lateral movement.

78. How does cyclic control create different amounts of lift within the rotor plane

The blade changes its angle of incidence during each revolution

Cyclic changes the incidence per revolution, giving different lift around the disc.

79. When flying straight ahead, where does the blade reach its minimum angle of incidence

Right in the front

Minimum incidence is right in the front.

80. When flying straight ahead, how many degrees after the minimum does the blade reach its maximum angle of incidence?

180°

The maximum is 180° later, opposite to the direction of flight.

81. In straight-ahead flight the blade has minimum incidence in front and maximum incidence at the rear. What is the result for the rotor plane?

It tilts forward, giving lift and forward thrust

More lift at the rear and less at the front tilts the plane forward, producing lift and forward thrust.

82. The pilot moves the cyclic stick backward. In which direction does the helicopter fly

Backward

Backward cyclic tilts the plane back and the helicopter flies backward.

83. What does the pilot do to the main rotor blades by moving the collective lever upwards

Increases the pitch angle of all blades at the same time

Collective up raises the pitch of all blades together.

84. The pilot moves the collective lever upwards. What does the helicopter do

It climbs

More pitch gives more lift and the helicopter climbs.

85. The pilot moves the collective lever downwards. What does the helicopter do

It descends

Collective down makes the helicopter descend.

86. Which hinges do the rotor blades need to change their pitch angle

Feathering hinges

Pitch change needs feathering hinges.

87. Where are the feathering hinges mounted

Between the rotor blade and the rotor head

Feathering hinges sit between blade and rotor head.

88. Which parts move the rotor blades to larger pitch angles when the collective is raised

Control rods

Control rods move the blades.

89. Besides changing the blade pitch, which other effect does a movement of the collective lever have?

It changes engine power and causes effects like torque

Collective movement also changes engine power and torque effects.

90. A pilot raises the collective lever to climb. The engine power changes and so does the torque effect on the fuselage. Which control must then vary the tail rotor thrust?

The tail rotor pedals

Torque is compensated by the tail rotor, whose thrust the pilot varies with the pedals.

91. Which control does the pilot use to turn the helicopter around the rotor mast axis

The tail rotor pedals

Pedals turn the helicopter around the mast axis.

92. What do the pedals change

The pitch angle of the tail rotor blades

Pedals change tail rotor blade pitch.

93. How do the pedals vary the tail rotor thrust

By changing the pitch angle of the tail rotor blades

Tail rotor thrust is varied via blade pitch.

94. The pilot moves the left pedal forward. What happens to the right pedal and the helicopter

Right pedal moves backward; helicopter turns left

Left pedal forward turns the helicopter left, with the right pedal moving back.

95. The pilot wants to turn the helicopter to the right around the rotor mast axis. What must he do?

Push the right pedal forward

Right pedal forward turns the helicopter right.

96. How is the collective pitch change achieved at the swash plate

By lifting or lowering the swash plate

Collective raises or lowers the swash plate.

97. How is the cyclic pitch change achieved at the swash plate

By tilting the swash plate

Cyclic tilts the swash plate.

98. A technician watches the swash plate while a helper moves one cockpit control, and the whole swash plate moves up without tilting. Which control is being moved?

The collective lever

Lifting the swash plate is collective; tilting would be cyclic.

99. During a control check the swash plate tilts but its mean height does not change. Which pitch change is being made?

Cyclic pitch change

Tilting the swash plate produces cyclic pitch change.

100. Where is the swash plate usually located

Underneath the rotor plane

It is usually underneath the rotor plane.

101. To which part of the swash plate are the rotor blade control rods connected

The upper part

Blade control rods attach to the upper part.

102. Why does the swash plate have to act as an interface

It transfers control signals between the stationary input system and the rotating rotor

Cockpit inputs are stationary but the blades rotate, so the swash plate links the two.

103. Which statement correctly assigns the swash plate parts?

Upper part rotating, lower part stationary

The upper part rotates; the lower part is stationary.

104. How are the rotating and stationary parts of the swash plate connected

By a bearing

A bearing connects the two parts.

105. How are the two parts of the swash plate arranged relative to each other

Parallel to each other

The two parts are parallel.

106. Which parts act on the stationary part of the swash plate

Control rods and bell cranks connected with the cockpit controls

The stationary part is moved by control rods and bell cranks from the cockpit.

107. How is the rotating part of the swash plate connected to the main rotor blades

By control rods

Control rods connect the rotating part to the blades.

108. What is the direct result on the rotor blades when the swash plate moves?

The control rods move and the blade pitch angle changes

Swash plate motion moves control rods, which change pitch.

109. Of which two parts does the swash plate consist

A rotating part and a stationary part

The swash plate has a rotating and a stationary part.

110. In a hover with no wind, how does the lift of the rotor blades compare

All blades generate the same amount of lift

In a no-wind hover all blades have the same parameters.

111. What is dissymmetry of lift

The difference in lift between advancing and retreating blade

The lift difference between advancing and retreating blade is dissymmetry of lift.

112. What is the crucial factor causing dissymmetry of lift when the helicopter flies

The airstream caused by the speed of flight

The flight airstream adds to or subtracts from blade speed.

113. Why does the advancing blade produce more lift in forward flight

The opposing airstream is added to its rotational speed

The airstream adds to its speed, giving more lift.

114. Why does the retreating blade produce less lift in forward flight

The opposing airstream is subtracted from its rotational speed

The airstream subtracts from its speed, giving less lift.

115. How does the speed change along the blade from root to tip

It increases towards the blade tip

Speed increases toward the tip.

116. In the manual example the helicopter flies forward at 100 kts. How does the relative wind change on the advancing and retreating blades?

Advancing +100 kts, retreating -100 kts

Flight speed adds on the advancing side and subtracts on the retreating side.

117. In the figure 'Dissymmetry of Lift' in forward flight, on which side of the helicopter is the advancing half?

The right side

The figure labels the right side as the advancing half.

118. A mass point on a rotating disk is moved towards the edge of the disk. What happens to the rotational speed and why?

It decelerates, because the mass resists the tangential acceleration

Moving outward needs higher circumferential speed; inertia resists and slows the rotation.

119. A mass point on a rotating disk is moved towards the centre of rotation. What does it do to the rotational speed?

It tries to accelerate the rotational speed

Moving inward, it keeps its higher speed and so accelerates the rotation.

120. Where does the blade's centre of gravity move when the blade flaps upwards

Closer towards the rotational axis

Upward flapping brings the CG closer to the axis.

121. Where does the blade's centre of gravity move when the blade flaps downwards

Closer to the blade tip

Downward flapping moves the CG outward toward the tip.

122. With constant rotor RPM, what does an upward flapping blade do

It accelerates and moves forward

Its CG moves inward, so it speeds up and leads.

123. With constant rotor RPM, what does a downward flapping blade do

It decelerates and moves backward

Its CG moves outward, so it slows down and lags.

124. Why is the installation of a lead-lag hinge required

Flapping blades accelerate or decelerate and move forward or backward

The forward and backward blade movement caused by flapping must be allowed by a lead-lag hinge.

125. In forward flight the advancing blade flaps up. Combining the flapping and centre-of-mass facts, how does this blade move in the plane of rotation?

It accelerates and moves forward (leads)

The advancing blade flaps up, its CG moves inward, so it accelerates and moves forward.

126. In forward flight the retreating blade flaps down. Combining the flapping and centre-of-mass facts, how does this blade move in the plane of rotation?

It decelerates and moves backward (lags)

The retreating blade flaps down, its CG moves outward, so it decelerates and lags.

127. Which self-correcting effect comes along with the dissymmetry of lift

The flapping of the blades

Blade flapping is the self-correcting effect.

128. Why does the advancing blade flap up

It experiences increased relative wind and generates more lift

More relative wind means more lift, so it flaps up.

129. What happens to the angle of attack as the advancing blade flaps up

It decreases

Upward flapping decreases the angle of attack.

130. What happens to the angle of attack as the retreating blade flaps down

It increases

Downward flapping increases the angle of attack.

131. The retreating blade experiences a decreased relative wind. Which sequence follows according to the manual?

Less lift, blade flaps down, angle of attack and lift increase

The retreating blade flaps down, which raises angle of attack and lift.

132. The advancing blade flaps up in forward flight. What finally happens to its lift?

The lift decreases as the angle of attack decreases

Upward flapping reduces angle of attack and so lift.

133. Is blade flapping alone sufficient to restore rotor balance

No, it helps to balance but is not sufficient

Flapping helps but does not fully restore balance.

134. Why does upward flapping reduce the angle of attack of the advancing blade, using the relative wind rule?

Moving up adds a downward relative wind component

Relative wind is opposite to blade motion, so an upward-moving blade meets air from above, reducing the angle of attack.

135. In the figure 'Blade Flapping', what does upward flapping do to the relative wind and angle of attack?

Changes the relative wind and decreases the angle of attack

The figure states upward flapping changes the relative wind and decreases the angle of attack.

136. In the figure 'Blade Flapping', a blade flaps downward. From which direction does the relative wind now meet the blade, and what is the result?

From below; the angle of attack increases

A blade moving down gets relative wind from below, which increases the angle of attack.

137. Why must a rotor blade be able to move and occupy a new position

Because aerodynamic forces act on the rotor blade

Aerodynamic forces require the blade to move.

138. Which two ways are given to achieve the flexibility for blade flapping

A flapping hinge or blade properties that absorb the forces

Either a flapping hinge or blades that absorb the forces.

139. Blade flapping compensates dissymmetry of lift. Which hinge lets the blade move up and down for this?

Flapping hinge

Flapping hinges allow the up-down movement.

140. When are lead-lag movements always generated

When a rotor blade is flapping

Flapping always generates lead-lag movement.

141. Which two causes of blade flapping lead to lead-lag movements

Dissymmetry of lift or cyclic control inputs

The manual names dissymmetry of lift and cyclic inputs.

142. A pilot makes a cyclic control input. Which chain of effects follows according to the manual?

The rotor plane tilts, which generates a lead-lag movement

Cyclic tilts the plane, and that generates lead-lag.

143. Which turning force does a counterclockwise turning main rotor generate on the fuselage

A clockwise turning force

The fuselage force is opposite, so clockwise.

144. During hover flight, which two jobs does the tail rotor thrust perform

Compensate main rotor torque and maintain heading control

Tail rotor thrust compensates torque and maintains heading.

145. What is translating tendency

The drift caused by the thrust of the tail rotor

Translating tendency is the tail-rotor-thrust drift.

146. How may the pilot prevent the lateral drift due to translating tendency

Tilt the main rotor disk slightly opposite to the drift

The disk is tilted slightly opposite to the drift.

147. A helicopter in hover drifts sideways although torque is compensated. What causes this drift?

The thrust of the tail rotor

The tail rotor's sideways thrust causes translating tendency.

148. During hover, tail rotor thrust makes the helicopter drift to the right. How must the main rotor disk be tilted to prevent this?

Slightly to the left

The disk is tilted opposite to the drift, so to the left for a right drift.

149. Which rotor thrust both compensates torque and causes translating tendency

The tail rotor thrust

The same tail rotor thrust compensates torque and causes the drift.

150. What is ground effect

A condition of improved performance when hovering near the ground

Ground effect improves performance near the ground.

151. What causes ground effect

Interference of the surface with the rotor airflow pattern

The ground surface interferes with the rotor airflow.

152. How does ground effect change as the helicopter gets closer to the ground

It becomes more pronounced

It is more pronounced nearer the ground.

153. Within which height is a helicopter operating in ground effect

Within one rotor diameter

In-ground effect is within one rotor diameter.

154. How does the power required to hover in ground effect compare with hovering out of ground effect?

Less power is required in ground effect

Hovering in ground effect needs less power.

155. A helicopter hovering out of ground effect descends into an in-ground effect hover. What happens to blade efficiency and why?

It increases, because the reduced downward airflow increases the angle of attack

Reduced downwash raises angle of attack and efficiency.

156. In ground effect, what happens to the lift vector

It becomes more vertical

The manual says the lift vector becomes more vertical.

157. In ground effect, what happens to the downward airflow velocity

It is reduced

Downwash velocity is reduced.

158. A pilot descends from an out-of-ground effect hover into ground effect and wants to remain in the ground effect hover. What must he do and why?

Reduce the blade pitch angle, because rotor efficiency increases

Entering ground effect raises angle of attack and efficiency, so pitch is reduced to stay in the hover.

159. A helicopter hovers in ground effect and the pilot climbs out of ground effect without changing the collective. Based on the text, what happens to the power requirement to keep hovering?

It increases

Hovering out of ground effect needs higher power than in ground effect.

160. During powered flight, how is the rotor drag overcome

With engine power

In powered flight, engine power overcomes rotor drag.

161. In which two failure cases must some other force sustain the rotor RPM

Engine failure or broken drive shafts

Engine failure or broken drive shafts require another force.

162. In which direction does the airflow pass through the main rotor during an autorotative descent

From bottom to top

In autorotation air flows up through the rotor.

163. What provides the energy to overcome blade drag and turn the rotor during autorotation

The airflow passing up through the rotor during descent

Upward airflow during descent drives the rotor.

164. How is the rotor driving force controlled in autorotation

By adjusting the collective pitch

The collective pitch controls it.

165. What is the 1st phase of autorotation

The controlled descent with the rotor driven by the airflow

The 1st phase is the controlled autorotative descent.

166. What does the pilot do in the 2nd phase of autorotation

Uses the rotor kinetic energy to cushion the touchdown

In the 2nd phase the stored kinetic energy cushions the touchdown.

167. In a controlled autorotative descent, which energy is converted to turn the rotor

Potential energy due to altitude

Potential energy from altitude drives the rotor.

168. As altitude decreases in autorotation, what happens to the energy stored in the turning rotor?

More potential energy is converted into kinetic energy stored in the rotor

Descent converts potential into kinetic energy stored in the rotor.

169. Why must the rotor RPM be sustained after an engine failure

So that a controlled flight can be continued to the ground

Rotor RPM keeps the helicopter controllable to the ground.

170. An engine fails in cruise. In which order does the pilot use energy during the autorotation?

Altitude drives the rotor in the descent, then rotor kinetic energy cushions touchdown

Phase 1 converts altitude into rotor energy; phase 2 uses the rotor energy for touchdown.